

How to run Guide:

Prerequisites:

- Python: 3.8 or higher
- OS: Windows, Mac, or Linux
- Hardware: CPU is supported (and default). A GPU (NVIDIA CUDA) is recommended for faster training but not required.

Setup & Installation

Windows:

```
python -m venv venv
.\venv\Scripts\activate
```

linux or mac:

```
python3 -m venv venv
source venv/bin/activate
```

Installation

run this command

```
pip install -r requirements.txt
```

Download the dataset

so will be keeping the dataset in this file so no need to download additionally

Model Training

Note: So this step is not necessary as i will be uploading the model which is pretrained , but in case if curious , then these are the following step's :

- Run the fine-tuning script. This will download the pre-trained ResNet18 and adapt it to the 7 emotion classes.

```
python finetune_train.py
```

- Output: Creates resnet_finetuned.pth (The trained weights). (Stage 1 deliverable)

Evaluation of the model

- Run this command to get other deliverable of stage 1 that is **confusion_matrix.png ,test_predictions.csv (Used by Stage 2 for testing)**

Also rag can be tested using `rag.py` on the terminal

Working Web app

This is the main product. It integrates the Vision Model and the RAG Chatbot into a single UI.

Run the app using command: `streamlit run app.py`

Troubleshooting

- "No module named..." errors Make sure your virtual environment is active ((venv) should appear in your terminal).

Run `pip install -r requirements.txt` again.

- Model Download Slowness The first time you run `app.py`, it downloads the `flan-t5-large` model (~1GB). This happens only once. Please be patient.